



LOB 1003 - Cálculo 1

Lista de exercícios 2 - Parte 1

1. $y = -\frac{1}{4}x + 1$

2. $y = -2x + 3$

3. $y = \frac{1}{3}x + \frac{2}{3}$

4.

a) $f'(x) = 6x$

b) $f'(x) = 9x^2 - 4x$

c) $f'(x) = -\frac{6}{x^3}$

d) $f'(x) = 3 - \frac{1}{x^2}$

e) $f'(x) = 2x^2 + \frac{x}{2}$

f) $f'(x) = 2 - x^{-2} - 2x^{-3}$

g) $f'(x) = 3x^2 + 2x$

h) $f'(x) = 3 + \frac{1}{2\sqrt{x}}$

i) $f'(x) = \frac{2}{3\sqrt[3]{x^2}}$

j) $f'(x) = -4x^{-2} - 10x^{-3}$

k) $f'(x) = \frac{1}{3\sqrt[3]{x^2}} + \frac{1}{2\sqrt{x}}$

l) $f(x) = 18x^2 + \frac{1}{3\sqrt[3]{x^2}}$

5.

a) $f'(x) = 6x - 5 \sin x$

b) $f'(x) = \sin x + x \cos x$

c) $f'(x) = \frac{\tan x - (x+1) \sec^2 x}{\tan^2 x}$

d) $f'(x) = \frac{\sec x [(3x+2) \tan x - 3]}{(3x+2)^2}$

e) $f'(x) = \sec x \left[\frac{1}{2\sqrt{x}} + \sqrt{x} \tan x \right]$

f) $f'(x) = \cotg x - x \operatorname{cosec}^2 x$

g) $f'(x) = 2x + 3 \tan x + 3x \sec^2 x$

h) $f'(x) = \frac{-\sin x - x(x+1) \cos x}{x^2 \sin^2 x}$

i) $f'(x) = \operatorname{cosec} x \left(3x^2 + \frac{1}{2\sqrt{x}} - (x^3 + \sqrt{x}) \cotg x \right)$

j) $f'(x) = \frac{-(x^2+1) \sin x - 2x \cos x}{(x^2+1)^2}$

$$\text{k) } f'(x) = 2x \operatorname{tg} x + x^2 \sec^2 x$$

$$\text{l) } f'(x) = \frac{-3(\cos x - \operatorname{sen} x)}{(\operatorname{sen} x + \cos x)^2}$$

$$\text{m) } f'(x) = \operatorname{sen} x(2x - 1) + (x^2 + 1) \cos x$$

$$\text{n) } f'(x) = -3 \operatorname{sen} x + 5 \operatorname{tg} x \sec x$$

$$\text{o) } f'(x) = 4 \operatorname{tg} x \sec x - \operatorname{cosec}^2 x$$

$$\text{p) } f'(x) = \frac{2x - (x^2 + 1) \operatorname{tg} x}{\sec x}$$

$$\text{q) } f'(x) = \frac{1 + x \cotg x}{\operatorname{cosec} x}$$

$$\text{r) } f'(x) = \frac{\cos x(x - 1) - \operatorname{sen} x(x + 1) - 1}{(x - \cos x)^2}$$

6. Calcule a derivada para cada uma das funções abaixo.

$$\text{a) } y' = (x^2 + 2x)e^x$$

$$\text{b) } y' = e^x(\cos x - \operatorname{sen} x)$$

$$\text{c) } y' = x(2 \ln x + 1) + 2e^x$$

$$\text{d) } y' = 5x(2 \ln x + 1)$$

$$\text{e) } y' = \frac{1 - \ln x}{x^2}$$

$$\text{f) } y' = 3 + \frac{5}{x}$$

$$\text{g) } y' = \frac{2e^x}{(1 - e^x)^2}$$

$$\text{h) } y' = \frac{-x - \ln x - 1}{(x \ln x)^2}$$

$$\text{i) } y' = \frac{e^x(x - 1)^2}{(x^2 + 1)^2}$$

$$\text{j) } y' = \frac{xe^x}{(x + 1)^2}$$

7. Calcule a derivada

$$\text{a) } f'(x) = 5^x \ln 5 + \frac{1}{x \ln 3}$$

$$\text{b) } g'(x) = 2 \cdot 3^{2x+1} \ln 3 + \frac{2x}{(x^1 + 1) \ln 2}$$

$$\text{c) } y' = 2(2^{x^2} x \ln 2 + 3^{2x} \ln 3)$$

- d) $f'(x) = x^{\sin 3x} \left(3 \cos 3x \ln x + \frac{\sin 3x}{x} \right)$
- e) $f'(x) = x^x [(1 + \ln x) \sin x + \cos x]$
- f) $y' = (3 + \cos x)^x \left[\ln(3 + \cos x) - \frac{x \sin x}{3 + \cos x} \right]$
- g) $y' = (2 + \sin x)^{\cos 3x} \left[-3 \sin 3x \ln(2 + \sin x) + \frac{\cos x \cos 3x}{2 + \sin x} \right]$
- h) $y' = \pi x^{\pi-1} + \pi^x \ln \pi$
- i) $y' = \ln 10 (10^x + 10^{-x})$

8.

- a) $y' = -2x, y'' = -2$
- b) $s' = 15t^2 - 15t^4, y'' = 30t - 60t^3$
- c) $y' = 4x^2 - 1 + 2e^x, y'' = 8x + 2e^x$
- d) $w' = -6z^{-3} + z^{-2}, w'' = 18z^{-4} - 2z^{-3}$
- e) $p' = \frac{q}{6} + \frac{q^{-3}}{6} + q^{-5}, q'' = \frac{1}{6} - \frac{q^{-4}}{2} - 5q^{-6}$
- f) $r' = -\frac{2}{3}s^{-3} + \frac{5}{2}s^{-2}, r'' = 2s^{-4} - 5s^{-3}$
- g) $y' = 2x - \frac{7}{x^2}, y'' = 2 + \frac{14}{x^3}$
- h) $r' = 3\theta^{-4}, r'' = -12\theta^{-5}$
- i) $w' = 6ze^z + 3z^2e^z, w'' = 6e^z + 12ze^z + 3z^2e^z$
- j) $y' = 12x - 10 + 10x^{-3}, y'' = 12 - 30x^{-4}$

9. Calcular as derivadas das funções abaixo.

- a) $y' = 10(2x + 1)^4$
- b) $y' = \left(1 - \frac{x}{8} \right)^{-8}$
- c) $y' = 4 \left(\frac{x^2}{8} + x - \frac{1}{x} \right)^3 \left(\frac{x}{4} + 1 + \frac{1}{x^2} \right)$
- d) $y' = \operatorname{tg}(\operatorname{tg} x) \sec(\operatorname{tg} x) \sec^2 x$
- e) $y' = 3 \sin^2 x \cos x$
- f) $y' = -5e^{-5x}$
- g) $y' = -7e^{5-7x}$

$$\text{h) } p' = -\frac{1}{2\sqrt{3-t}}$$

$$\text{i) } s' = \frac{4}{\pi} \cos 3t - \frac{4}{\pi} \sin 5t$$

$$\text{j) } r' = \frac{\operatorname{cosec} \theta}{\cotg \theta + \operatorname{cosec} \theta}$$

$$\text{k) } y' = 2x \sin^4 x + 4x^2 \sin^3 x \cos x + \cos^{-2} x + 2x \cos^{-3} x \sin x$$

$$\text{l) } y' = e^{-x}(1-x) + 3e^{3x}$$

$$\text{m) } y' = \left(\frac{5}{2}x^2 - 3x + 3 \right) e^{\frac{5x}{2}}$$

$$\text{n) } f'(\theta) = \frac{2 \sin \theta}{(1 + \cos \theta)^2}$$

$$\text{o) } r' = 2\theta \cos(\theta^2) \cos(2\theta) - 2 \sin(\theta^2) \sin(2\theta)$$

$$\text{p) } y' = 2\theta e^{-\theta^2} \sin(e^{-\theta^2})$$

$$\text{q) } y' = -2 \sin(2t-5) \cos(\cos(2t-5))$$

$$\text{r) } y' = \frac{-t \sin(t^2)}{\sqrt{1 + \cos(t^2)}}$$